Test:	When does it apply:	What to do:
Divergence Test	Always.	Evaluate $\lim_{n\to\infty} a_n$ or $\lim_{n\to\infty} \lvert a_n \rvert$. → If $\lim_{n\to\infty} \neq 0$: • Divergent →If $\lim_{n\to\infty} = 0$: • No conclusion
Integral Test	If $\sum a_n = \sum F(n)$ for F(x), a positive, continuous, decreasing function on [a, ∞].	Solve $\int_a^\infty f(x)dx$. → Series converges if integral converges. → Series diverges if integral diverges.
Direct Comparison Test	If $\sum a_n$ contains only positive terms and there is a known series $\sum b_n$ that is similar.	Compare a_n vs. b_n sequences. → If $a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. → If $a_n \geq b_n$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.
Limit Comparison Test	Same as DCT.	Evaluate $\lim_{n\to\infty} \frac{a_n}{b_n}$. Assuming $0 < \lim_{n\to\infty} \frac{a_n}{b_n} < \infty$, then: → If $\sum b_n$ converges, then $\sum a_n$ converges. → If $\sum b_n$ diverges, then $\sum a_n$ diverges.
Alternating Series Test	Only when $\sum a_n$ is alternating.	Check $\lim_{n\to\infty} \lvert a_n \rvert = 0$. Check $\lvert a_n \rvert$ is decreasing. → If so, series converges.
Ratio Test	Always: Typically apply when a_n has factorials or other "fast" growing parts.	Evaluate $\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$. → If lim < 1: • Absolutely Convergent → If lim > 1: • Divergent → If lim = 1: • Inconclusive
Root Test	Always: Typically apply when $a_n = (b_n)^n$ or similar.	Evaluate $\lim_{n\to\infty} \lvert a_n \rvert^{1/n}$. → If lim < 1: • Absolutely Convergent → If lim > 1: • Divergent → If lim = 1: • Inconclusive

Special Series:

- Geometric: $a + ar + ar^2 + \ldots = \sum_{n=1}^{\infty} ar^n$
  - Partial Sum Formula: $S_k = a + ar + ar^k = \frac{a - ar^{k+1}}{1-r}$
  - Series Formula: $\sum_{n=0}^{\infty} ar^n = \begin{cases} \frac{a}{1-r}, & |r| < 1 \\ Divergent, & |r| \geq 1 \end{cases}$
- Telescoping: "Most" terms a_n will "cancel out" when determining SOPS (S_k).
  - Can get simple formula for S_k.
  - Take limit to evaluate series.
- **p**-series: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ or $\sum_{n=1}^{\infty} \frac{C}{n^p}$ for positive constant C.
  - Convergent if $p > 1$
  - Divergent if $p \leq 1$

General Information:

- Terms and Definitions
  - Sequences of terms → a_n
  - Sequences of Partial Sums (SOPS)
    - $S_k = a_1 + \ldots + a_k = \sum_{n=1}^{k} a_n$
  - Series → $S = \sum_{n=1}^{\infty} a_n$
- Testing with Special Series
  - Use Direct Comparison (DCT) and Limit Comparison (LCT) to compare given series with p-series and geometric series since we easily know when these two types of series converge or diverge.
- Absolute vs. Conditional Convergence
  - Harder to be Absolutely Convergent because all terms are positive and still convergent.
  - Absolute Convergence means the alternating series is certainly convergent but Conditional Convergence doesn't necessarily mean that the alternating series is absolutely convergent.
  - Absolute Convergence is determined:
    - With any test of $\sum a_n$, but as $|a_n|$.
    - Ratio Test
    - Root Test
  - Alternating Series Test (AST) will determine Conditionally Convergence.
- Alternation Series Remainder Theorem
  - How to bound the difference between full series $S = \sum a_n$ and partial sum $S_k = \sum a_n$ when a_n is alternation: $|S - S_k| < |a_{k+1}|$.